简体中文 | English

Tongxin **ZHOU**

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Hardware/software co-design for efficient AI systems | accelerator architecture | FPGA/RTL prototyping

Research Profile

- **Overview**: I am a first-year Master's student in Integrated Circuit at *Nanjing University*, with a Bachelor's degree in Microelectronic from *Sun Yat-sen University*. My research centers on **hardware-software co-design** for efficient AI systems, with a core focus on bridging the gap between **LLM/foundation model algorithm advances** and practical **hardware deployment**. I validate research ideas through **RTL-driven FPGA prototyping**, targeting **high-performance, energy-efficient AI inference** for **resource-constrained scenarios**, and am pursuing a PhD in **computer architecture** and **AI accelerator design**.
- **Research Focus**: AI Accelerator Architecture & Hardware-Software Co-Design; Hardware-Aware LLM Compression & Efficient Inference; Reconfigurable Computing & Network-on-Chips(NoC); Memory Hierarchy & Dataflow Optimization for AI Workloads.
- **Research Questions**:
  - How to balance **architectural programmability** and **domain-specific specialization** for rapidly evolving foundation models?
  - How to jointly optimize **model compression**, **dataflow mapping**, and **hardware design** for measurable real-world inference gains?
  - How to break through **memory/bandwidth bottlenecks** of LLM inference under tight edge-side latency and energy constraints?
- **Technical Strengths**: RTL Design & FPGA Prototyping (Verilog/SystemVerilog, Xilinx Vivado); Hardware-Aware Quantization, Structured Sparsity & Pruning; Digital IC Design; AI Model Deployment & Hardware-Software Co-Verification.

Education

Nanjing University, School of Integrated Circuits
Suzhou, Jiangsu, P.R. China
September 2025 – June 2028 (Expected)

M.S. in Integrated Circuit Science and Engineering **(Recommendation)**
Advisor: Assoc. Prof. **Yuxiang Fu** (website) | **GPA**: 4.0/5.0
Selected Coursework: Semiconductor Materials Science (97), Principles of Computer Organization (95), Digital Integrated Circuits (90)

Sun Yat-sen University, School of Integrated Circuits
Shenzhen, Guangdong, P.R. China
September 2021 – June 2025

B.Eng. in Microelectronic Science and Engineering
GPA: 86.17/100 | **Rank**: 7/87 (Top 8%)
Undergraduate Thesis: Analysis of Merging Algorithms for Domestic Large Language Models - Awarded Excellent Graduation Thesis (top 20%)
Selected Coursework: Artificial Intelligence and Hardware Acceleration (91.5), Hardware Description Language Design (91), Electronic Design Automation (93)

Publications & Manuscripts

- **Chenyang Dai, Yijin Liu, Tongxin Zhou, Yuxiang Fu*. ARgent: A Multi-Agent-Driven Iterative Framework for Automatic Approximate Circuit Generation. Under Review at IEEE/ACM 2026 International Conference on Computer-Aided Design (ICCAD), 2025.**
  - Contributed to the **multi-agent** framework design for automated **approximate circuit** generation under user-specified constraints;
  - Achieved near-optimal energy efficiency and superior circuit quality compared to existing manual and automated design methods.
- **李沛鸿, 贺傍, 周彤昕, 傅玉祥*. *基于输出激活分布的神经网络动态量化方法*. 南京大学学报, 2026.**
  - Investigated data distribution characteristics of MoE LLMs, and raised stage-aware quantization strategies for **prefill** and **decode** phases;
  - Realized an improved **efficiency-accuracy trade-off** for edge LLM inference via progressive quantization configuration search.

Research Experience

Academic Project

Software-Hardware Co-Optimization for Edge Foundation Models

VLSI Lab, Nanjing University | Advisor: Assoc. Prof. **Yuxiang Fu** | Jan 2025 – Present

1. Investigate data distributions in MoE LLMs and implement SOTA quantization methods (e.g., SmoothQuant and GPTQ) on MoE models.
2. Design the compute and memory modules for masked Attention, covering the full dataflow from storage to on-chip execution.
3. Lead instruction-level testing and on-board debugging after FPGA deployment, validating end-to-end read/write correctness and iteratively refining RTL until stable operation.

Construction and Optimization of a Chinese Large Language Model Based on Parameter Fusion

School of Integrated Circuits, Sun Yat-sen University | Advisor: Assist. Prof. **Siyu Liao** | Sep 2024 – Jun 2025

1. Proposed MELT, a cross-architecture LLM fusion framework combining architecture adaptation, sensitivity-aware quantization, and lightweight routing for domain-oriented deployment.
2. Built a compatibility evaluation and preprocessing pipeline for heterogeneous Chinese LLMs, including layer alignment and dynamic vocabulary expansion to improve cross-architecture merging feasibility.
3. Developed a hierarchical hybrid-precision quantization method that kept top-[K] sensitive layers in FP16 and optimized the remaining layers with low-bit quantization, reducing model size by 37.5% while improving mathematical-task accuracy over INT8 baselines.
4. Completed the undergraduate thesis based on this work and received Excellent Graduation Thesis (Top 20%).

Design of a Depth Estimation Hardware Processor for Robotic Vision Systems

Undergraduate Innovation Program, Sun Yat-sen University | Wendong Mao (in the team of **Zhongfeng Wang**) | Sep 2023 – Jun 2024

1. Built an end-to-end FPGA-based depth estimation system that captured images from an OV7670 camera and displayed real-time monocular depth results on screen.
2. Implemented and optimized a ResNet-based depth estimation pipeline, integrating fast CNN techniques and hardware-oriented model processing for real-time inference.
3. Developed and validated the full on-board prototype, including sensor input, model deployment, hardware integration, and real-time output visualization.

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Research Experience

Academic Projects

Low-Power High-Resolution Delta-Sigma Modulator for MEMS Capacitive Accelerometers

Undergraduate Innovation Program, Sun Yat-sen University | Asso. Prof. **Quanzhen Duan** | Mar 2022 – Dec 2022 | Team Lead

1. Led the design of a delta-sigma ADC for MEMS accelerometers, targeting high-resolution sensing under microwatt-level power.
2. Completed end-to-end parameter modeling, circuit design, simulation verification, technical documentation, and project defense.
3. Optimized the circuit architecture to achieve **16-bit** effective resolution, **1.67 μW** power consumption, and **101 dB** quantization SNR, and delivered a fully validated design with complete reports and simulation results.

Competition Projects

LAV: Design of a LoongArch-Based Accelerator for R-MSFM Models

8th National College Student IC Innovation and Entrepreneurship Competition | Core Contributor | Mar 2024 – Aug 2024

1. Proposed a highly customized hardware accelerator architecture for real-time visual inference based on the LoongArch ISA;
2. Led RTL coding, circuit design, FPGA board bring-up, simulation verification, and on-site live demonstration for the competition;
3. Optimized accelerator dataflow and memory hierarchy to improve throughput for edge vision workloads;
4. Awarded **Second Prize (South China Region)** in the national-level competition (top 8% of participating teams).

Motion Target Control and Tracking System

National Undergraduate Electronic Design Contest | Core Contributor | Aug 2023

1. Designed a button-controlled embedded system for motion target reset, winding, path following, and real-time tracking;
2. Completed core programming, circuit schematic design, PCB layout, and technical report writing for the competition;
3. Implemented all required functions with stable performance, passing full on-site validation at the competition venue;
4. Awarded **Second Prize (Guangdong Division)** in the national-level electronic design contest.

Chemical Composition Analysis Model for Ancient Glassware Based on K-Means Clustering

National Mathematical Modeling Contest | Team Lead | Oct 2022

1. Led a 3-person team to build a data analysis model for ancient glassware chemical composition, addressing the core problem;
2. Completed principle research, algorithm design, code implementation, and full contest report drafting as team lead;
3. Optimized the K-Means clustering model to achieve high classification accuracy for glass type and weathering degree;
4. Awarded **Second Prize (Guangdong Division)** in the national-level mathematical modeling contest.

Selected Honors & Awards

First-Class Outstanding Student Scholarship, Nanjing University	November 2025	Nanjing University
Outstanding Graduate, School of Integrated Circuits	June 2025	Sun Yat-sen University
Second Prize (South China Region), National College Student IC Innovation & Entrepreneurship Competition Engineering Design Contest	August 2024	Ministry of Education of P.R. China
National Scholarship (China) **[TOP 1 of 87 in school]**	December 2023	Ministry of Education of P.R. China
First-Class Outstanding Student Scholarship	November 2023	Sun Yat-sen University
Second Prize (Guangdong Division), National Undergraduate Electronic Design Contest Engineering Design Contest	August 2023	Ministry of Education of P.R. China
Special Scholarship, Sun Yat-sen University	November 2022	Sun Yat-sen University
Second Prize (Guangdong Division), National Mathematical Modeling Contest Engineering Design Contest	October 2022	China Society for Industrial and Applied Mathematics

English Proficiency

TOEFL iBT: 80/120 (CEFR B2)	**CET-4: 550/710 | CET-6: 485/710**	**GRE: N/A**

Technical Skills

Hardware Design & FPGA	Programming & AI Tools	EDA & Simulation
Verilog/SystemVerilog, RTL Design, FPGA Prototyping, Digital IC Design	C, Python, PyTorch, MATLAB, Linux, Git, Inference (vLLM, llama.cpp), Compression (AWQ, GPTQ, GGUF)	Cadence EDA Suite, Xilinx Vivado, Altium Designer, Modelsim Simulation